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Course : R-VIJETA (R-JP)

PHYSICS DPP
DAILY PRACTICE PROBLEM

REVISION DPP NO. # 02

Target : JEE (Advanced) 2023

DPP Syllabus
KINEMATICS, NLM, FRICTION, WPE, CURRENTELECTRICITY, CAPACITANCE, ERROR & MEASUREMENT
DPP NO. 2

Total Marks : 170	**Max. Time : 160**	
One or more than one may be correct option ('–1' Negative marking) Q.1 to 20;	**(3 Marks; 2 Min.)**	**[60, 40]**
Numerical based questions (No negative marking) Q.21 to Q.40;	**(4 Marks; 5 Min.)**	**[80, 100]**
Single Choice Objective ('–1' Negative marking) Q.41 to Q.50	**(3 Marks; 2 Min.)**	**[30, 20]**

1. A particle of mass 'm' moves under a conservative force with potential energy $U(x) = \frac{cx}{x^2 + a^2}$ where c and a are positive constants. Choose the correct option(s)

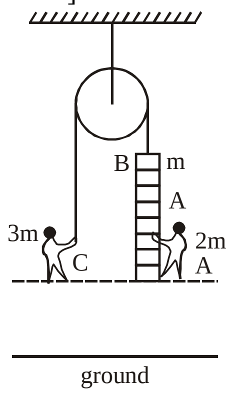
(A) Position of unstable equilibrium is $x = +a$

(B) The particle can oscillate about $x = -a$

(C) If particle's velocity is more than $\sqrt{\frac{c}{ma}}$ at origin in positive x-direction, then it will reach $x = +\infty$

(D) If particle's velocity is more than $\sqrt{\frac{c}{ma}}$ at $x = -a$ in negative x-direction, then it will reach $x = -\infty$

2. Man A of mass 2 m is standing on a ladder of mass m. Ladder is attached with an inextensible light string and man C of mass 3 m is holding the other end of string. Initially, the whole system is at rest. Assume whole system is at sufficient height and pulley is frictionless. Also A and C are at same horizontal level. Then choose the correct options. [$g = 10\ m/s^2$]



(A) If man A starts climbing up on the ladder with a constant acceleration of 4 m/s^2 w.r.t ground then, the vertical distance between A and C after 1 sec is 1 m.

(B) If man C starts climbing up on the rope (while A remain at rest with respect to ladder) with constant acceleration of 4 m/s^2 with respect to ground then, the vertical distance between A and C after 1 sec is 2 m.

(C) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 4 m/s^2 upward.

(D) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 3 m/s^2 upward.

3. A body is thrown from ground with an angle α returns to ground after covering a horizontal distance b. Then this body is projected with same velocity at same angle α above horizontal from a tower of height h_1 and returns to the ground after covering a horizontal distance s. If this body is projected with same speed at angle α (below the horizontal) from tower of height h_2 so that it will fall on the ground at the same distance s from this tower base, then choose the correct option(s) :

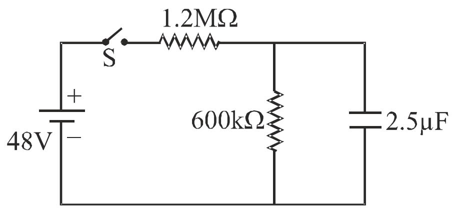
(A) $h_1 + h_2 = \dfrac{2s^2}{b}\tan\alpha$

(B) $h_2 - h_1 = 2s\cot\alpha$

(C) $\dfrac{h_2}{h_1} = \dfrac{s+b}{s-b}$

(D) $\dfrac{h_2}{h_1} = \dfrac{s}{2b}$

4. In the circuit shown in figure capacitor was initially uncharged, switch is closed at t = 0. Select correct alternatives.



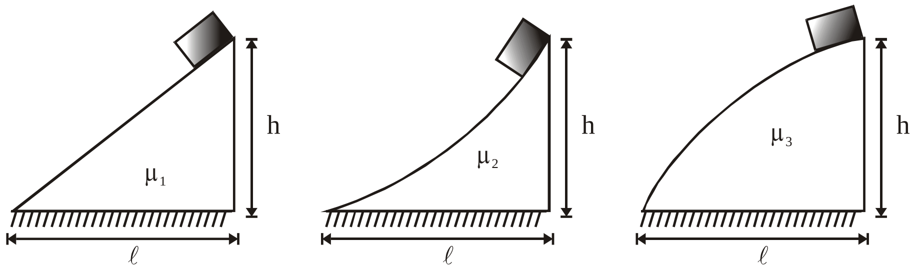
(A) The initial battery current immediately after switch S is closed is 40 μA

(B) The battery current, long time after switch S is closed is $\dfrac{80}{3}$ μA

(C) The time after which current through capacitor becomes half is $\ell n4$ sec.

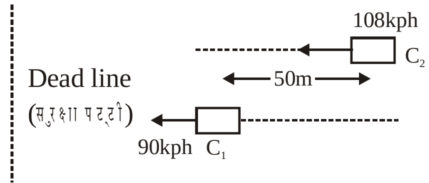
(D) The current through the 600 kΩ as a function of time is $\dfrac{80}{3}(1 - e^{-t})$ μA.

5. Consider three fixed surfaces shown in the figure. Three blocks each of mass m are released from rest from top of the three surfaces. All blocks reach ground with same speed. Length of path travelled by the blocks is same for second and third surface. If coefficient of friction of three surfaces are μ_1, μ_2 and μ_3 respectively then



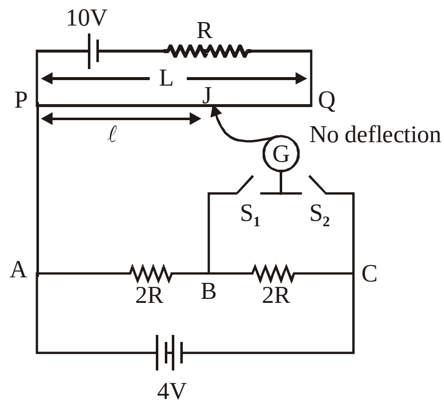
(A) $\mu_1 = \mu_2$ (B) $\mu_1 > \mu_2$ (C) $\mu_2 = \mu_3$ (D) $\mu_2 < \mu_3$

6. Two cars C_1 & C_2 are moving in parallel lanes in the same direction at speeds 90 km/hr & 108 km/hr respectively. As the traffic signal turns red at the instant shown in figure, both apply brakes(assume constant retardation) simultaneously. If they both stop together at the dead line:



(A) distance of dead line from C_2 is 300 m
(B) distance of dead line from C_1 is 250 m
(C) time taken by the cars to end up after the signal turn red is 15 sec
(D) time taken by the cars to end up after the signal turn red is 20 sec

7. Figure shows a potentiometer circuit. The length of potentiometer wire is L and its resistance is 4R. Cell is ideal, select correct option –



(A) when only S_1 is closed $\frac{\ell}{L} = \frac{1}{4}$ (B) when only S_2 is closed $\frac{\ell}{L} = \frac{1}{2}$

(C) When S_1 and S_2 both closed $\frac{\ell}{L} = \frac{1}{4}$ (D) When S_1 and S_2 both closed $\frac{\ell}{L} = \frac{1}{2}$

8. N particles moving in a straight line have initial velocities of 1, 2, 3, ... N m/s and accelerations of 1, 2, 3 ..... N m/s^2 respectively. If the initial spacing between any two consecutive particles is same, then select the correct alternative(s)

(A) the distance between any two particles remains constant throughout

(B) the distance between any two consecutive particles is same for all particles

(C) the distance between any two consecutive particles is different and increases with time

(D) the distance between any two consecutive particles increases with time

9. Which of the dimensional formula is/are correct ?

(A) [permittivity of vacuum × electric field strength] = $[M^0L^{-2}T^1A^1]$

(B) [electrical conductivity] = $[M^{-1}L^{-3}T^3A^1]$

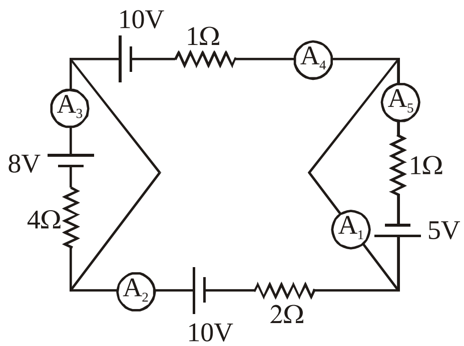
(C) [viscous force] = $[M^1L^1T^{-2}]$

(D) $\left[\dfrac{\text{electric field strength}}{\text{magnetic field induction}}\right] = [M^0\,L^1\,T^{-1}]$

10. Which of the following is/are conservative force(s) ?

(A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = \dfrac{5}{r}\hat{r}$ (C) $\vec{F} = \dfrac{3(x\hat{i} + y\hat{j})}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \dfrac{3(y\hat{i} + x\hat{j})}{(x^2+y^2)^{3/2}}$

11. In the given circuit all ammeters are ideal. Then choose correct statement.



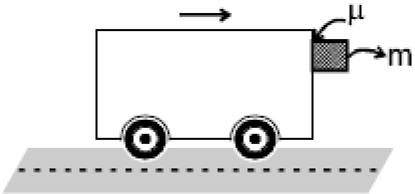
(A) Reading of A_2 = Reading of A_5.

(B) Reading of A_2 and Reading of A_4 both are zero.

(C) Reading of A_1 is 5 amp

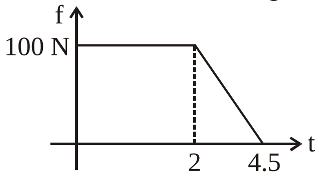
(D) Reading of A_1 = $\dfrac{5}{2}$ times reading of A_3.

12. The Van is moving on a horizontal ground along a straight road as shown in figure.



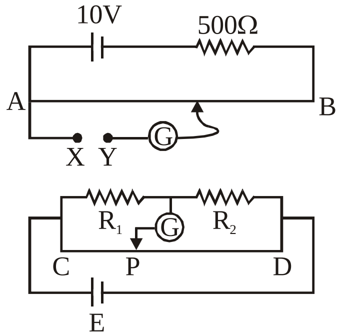
The velocity of Van is given as :
$v = (9t - t^2 + 10)$ m/s, where t is in second

Coefficient of friction between the wall of van and block of m kg is μ. The block was at rest w.r.t. van initially. The friction time graph between block and van is given as ($g = 10\ m/s^2$).



(A) The mass of block is 10 kg.
(B) Work done by normal reaction on van is first 5 sec is 202.5 J
(C) Coefficient of friction between van and block is 0.5.
(D) Acceleration of block at t = 3 sec is 5 m/s^2

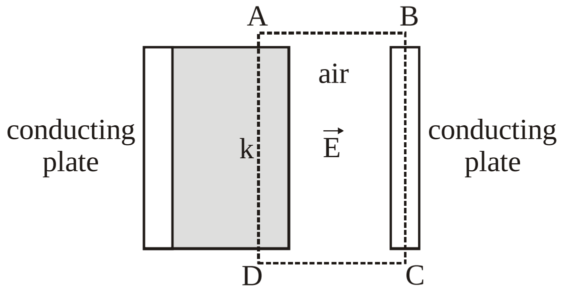
13. In the circuit shown, AB is potentiometer wire of length 10m and resistance 500 Ω. CD is a 1m wire of meter bridge which is balanced at CP = 20 cm. Given that when the potential difference across R_1 is applied across XY the balancing length on potentiometer is 2m and for potential difference across R_2, the corresponding length is 8 m. Then



(A) $\frac{R_1}{R_2} = \frac{1}{4}$ (B) Potential difference across R_1 is 1V
(C) Potential difference across R_2 is 4V (D) EMF of the battery E = 5V

14. Consider two sets of a measurement
Set-1 : 10.8 g, 10.7 g, 10.9 g
Set-2 : 10.431 g, 10.742g, 9.821 g
In set-1
(A) precision is greater in comparison to set-2
(B) precision is smaller in comparison to set-2
(C) Mean absolute error is greater comparison to set-2
(D) Mean absolute error is same in both sets

15. The figure shows two thin large identical parallel conducting plates placed in air with a slab of dielectric constant k placed between them. The width of the small gap between the conducting plates is d while the thickness of the dielectric slab is t. The conducting plates are given equal and opposite charges due to which the strength of the electric field in air region is E as shown. The area of plates perpendicular to the plane of figure is A and consider it to be large. Neglect the edge effects. (The portion ABCD has the same dimensions perpendicular to the plane of the figure as that of plate or dielectric medium) Mark the correct options :



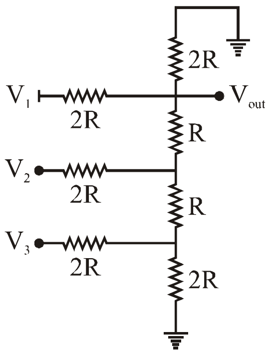
(A) The net charge in the portion ABCD (volume) is zero.

(B) The magnitude of net charge in the portion ABCD is $\frac{\varepsilon_0 EA}{k}$.

(C) The electrostatics energy stored in the dielectric medium is $\frac{\varepsilon_0 E^2 At}{2k}$.

(D) The electrostatic energy stored in the dielectric medium is $\frac{\varepsilon_0 E^2 At}{2k^2}$.

16. In the given circuit diagram V_1, V_2 and V_3 can take value either zero or 1V. V_{out} is output voltage. Which of the following is/are correct for V_{out}?



(A) If $V_1 = 1V$, $V_2 = 1V$, $V_3 = 1V$, then, $V_{out} = \frac{7}{12}V$.

(B) If $V_1 = 1V$, $V_2 = 0$, $V_3 = 1V$, then, $V_{out} = \frac{5}{12}V$.

(C) If $V_1 = 0$, $V_2 = 1V$, $V_3 = 1V$, then, $V_{out} = \frac{1}{4}V$.

(D) If $V_1 = 0$, $V_2 = 0$, $V_3 = 1V$, then, $V_{out} = \frac{1}{6}V$.

17. Height of a mountain at a place (h) is measured by measuring density of rock's material (d) forming the mountain, Young's Modulus of rock's material (Y) and acceleration due to gravity (g) at that place. Percentage error in measurement of d, Y and g are respectively + 2%, + 1% and + 2%, then

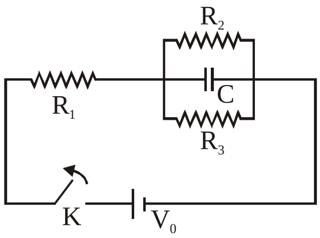
(A) $h \propto Y$

(B) $h \propto \frac{g}{d^2}$

(C) percentage error in measurement of h is + 5%

(D) percentage error in measurement of h is + 7%

18 In the circuit shown in the figure, switch K is kept closed for long time and then it is opened. The constant voltage across the terminals of the battery is $V_0 = 9$ V, the capacitance of the capacitor is C = 50 μF, the values of resistance R_1 and R_2 are equal, $R_1 = R_2 = 100\ \Omega$. Select correct statement for the process after opening the switch.



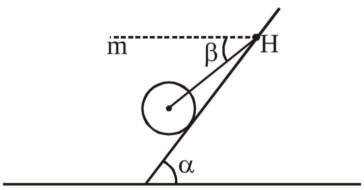
(A) If $R_3 = 400\ \Omega$, then charge on capacitor just after opening the switch is 200 μC.

(B) If $R_3 = 80\ \Omega$, then charge flown through R_3 till steady state is 40 μC.

(C) If $R_3 = 400\ \Omega$, then charge flown through R_2 till steady state is 160 μC.

(D) Charge flown through R_3 till steady state will be maximum if $R_3 = 50\sqrt{2}\ \Omega$.

19. A uniform solid sphere of mass 'm' rests on smooth incline with help of an ideal string, string is tied to point H on incline and attached to the centre of sphere, then ($\beta < \alpha$) :



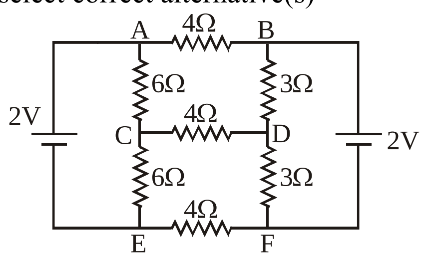
(A) tension in the string is $\frac{mg\sin\alpha}{\cos(\alpha-\beta)}$

(B) tension in the string is $\frac{mg\sin\alpha}{\cos\beta}$

(C) normal force on sphere is $\frac{mg\cos\beta}{\cos(\alpha-\beta)}$

(D) normal force on sphere is $\frac{mg\cos(\alpha-\beta)}{\cos\beta}$

20. Consider following circuit and select correct alternative(s)



(A) Current through each cell is $\frac{1}{6}$A .

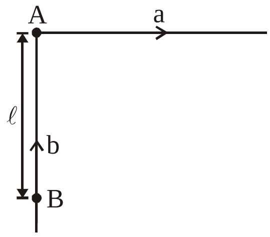
(B) Current through each 6 Ω resistance is $\frac{1}{6}$A .

(C) Current through each 3 Ω resistance is $\frac{1}{3}$A .

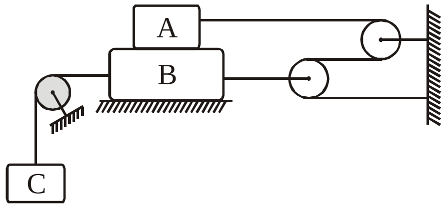
(D) Current through each 4 Ω resistance is $\frac{1}{2}$A .

21. A man in a boat wishes to cross a 400 m wide river flowing at the rate of 2 m/s. Speed of boat in still water is 4 m/s. The person starts to row the boat perpendicular to the river flow from one bank of the river. When he reaches the middle of the river, he starts to row the boat at an angle θ with the direction perpendicular to river flow so as to reach the point directly opposite to the starting point on the opposite bank. The value of $\tan\theta$ is $\frac{4}{n}$. Find value of n.

22. In the figure shown, A and B are two particles which start from rest. A has constant acceleration $a = 3\ m/s^2$ in the direction shown. B also increases its speed at a constant rate $b = 4\ m/s^2$, but the direction of velocity is always towards A. If the time after which B meets A is $\sqrt{\frac{2\ell b}{b^2+\alpha}}$ then, α (in SI unit) is :



23. Block A of weight 50 N and block B of weight 70 N are connected by rope pulley system as shown. The largest weight C that can be suspended without moving block A and B is W. The coefficient of friction for all plane surfaces of contact is 0.3. If the pulleys are ideal, then the value of W (in N) is .



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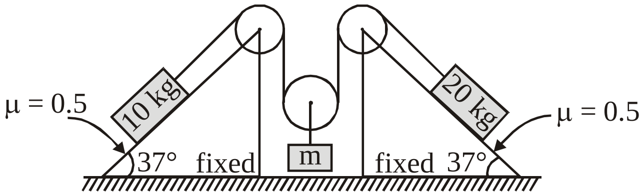
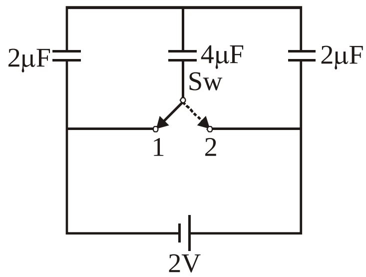
24. In given arrangement, 10 kg and 20 kg blocks are kept at rest on two fixed inclined planes. All strings and pulleys are ideal. The difference of maximum and minimum values of m (in Kg) for which system remain in equilibrium is : ($g = 10\ m/s^2$)

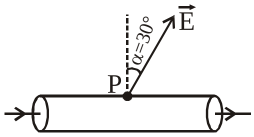


25. A particle is thrown with initial speed u = 1 m/s in vertically upward direction. Due to air resistance, retardation produced in the particle is proportional to square of velocity of particle i.e. αv^2, where α is $10\ m^{-1}$. If time taken by particle to reach maximum height is $\frac{\pi}{x}$ sec then, x is (Take $g = 10\ m/s^2$).

26. What amount of heat will be generated in the circuit shown in the figure, after the switch S*w* is shifted from position 1 to position 2?



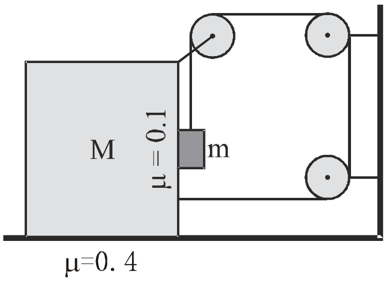
27. There is a long solid conductor of resistivity $\rho = 1 \times 10^{-6}\ \Omega$-m, surrounded by air. There is a steady electric current along the length of the conductor. At point 'P' just outside the cylinder the electric field strength $E = 10^{-4}$ V/m is directed at an angle of α to the normal to the surface. If the current density in the conductor in the vicinity of point 'P' (see fig.) is 10a, find a.



28. To measure the diameter of a wire, a screwgauge is used. In a complete rotation, spindle of the screw gauge advances by $\frac{1}{2}$ mm and its circular scale has 50 divisions. The main scale is graduated to $\frac{1}{2}$ mm. If the wire is put between the jaws, 8 main scale divisions are clearly visible and 10 divisions of circular scale coincide with the reference line. The resistance of the wire is measured to be ($10\ \Omega \pm 1.5\%$). Length of the wire is measured to be 10 cm using a scale of least count 1 mm. Maximum percentage error in resistivity measurement is : (in nearest integer)

29. Electric field, in a straight solid uniform cylindrical conductor of radius R = 2 cm is along the axis of the cylinder and is given by $E = Kr$ where $K = 1 \times 10^8$ Volt/Metre^2 and 'r' is the distance (in metre) from the axis of the cylinder. If current in the cylinder is $I = (16\pi)$ micro ampere. If the specific conductivity $\sigma = z \times 10^{-8}$. Find the value of z. [Read $\sigma = z \times 10^{-8}$ (in S.I Units.)]

30. In the system shown, M = 10 kg. The coefficient of friction between block and ground is 0.4 and that between two blocks is 0.1. The maximum value of m (in kg) so that the arrangement shown in the figure is in equilibrium is given by P/4. Find the value of P.



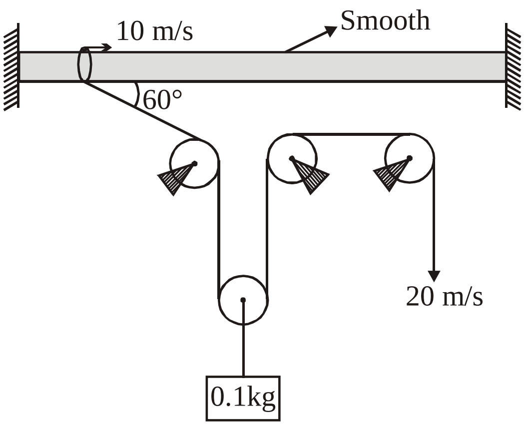
31. Two resistances are measured in Ohm.

$R_1 = 3\Omega \pm 1\%$

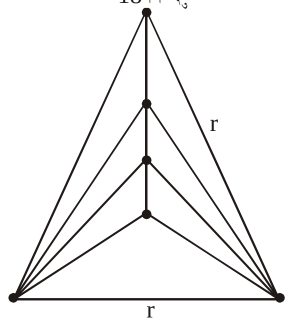
$R_2 = 6\Omega \pm 2\%$

When they are connected in parallel, maximum percentage error in equivalent resistance is α. Find 3α.

32. All the pulleys are ideal, string is massless then rate of work done by gravity at the given instant is 'X' Watt, then calculate 'X' :

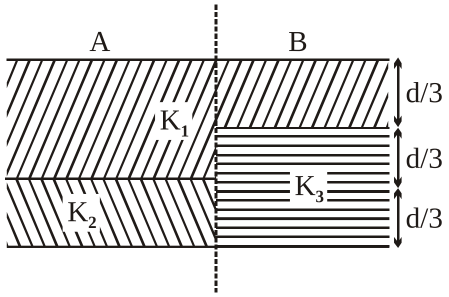


33. There are 20 points on a circuit board. Each point is connected to each of the other points by a wire of resistance r = 60 Ω. Find the equivalent resistance R (in Ω) between any two points.

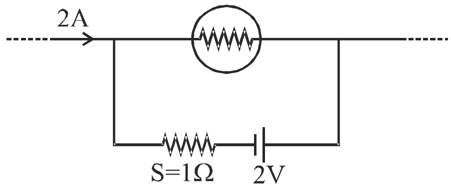


So equivalent resistance will be 6 Ω.

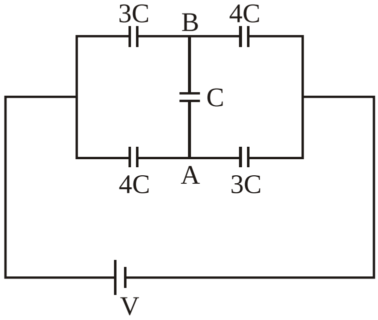
34. A capacitor is filled with 3 dielectric as shown figure. The value of dielectric constant K_1, K_2 and K_3 is 2, 4 and 6 respectively. If the ratio of surface charge density on left half to the right half of plate of capacitor to the simplest integer is $\alpha : \beta$, then $(\alpha + \beta)$ is -



35. The galvanometer shown in the figure has resistance 10 Ω .It is shunted by a series combination of a resistance S = 1 Ω and an ideal cell of emf 2 V. A current 2A passes as shown. The potential difference (in volts) across the resistance S is



36. Consider the given circuit. Initially all capacitor were uncharged.



If $V_A - V_B = 1$ volt in steady state, the potential difference across the terminals of battery is

37. In a vernier caliper, 10 vernier scale division coincides with 9 main scale division. Where 1 main scale division is 0.1 cm.

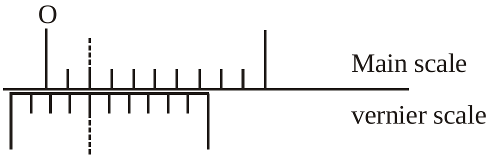
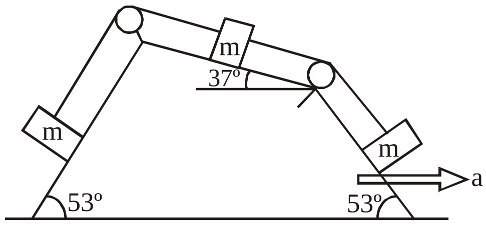


Figure shows main scale and vernier scale when gap between two jaws is zero. The zero error in the faulty vernier calliper is x (in mm), then 5x is

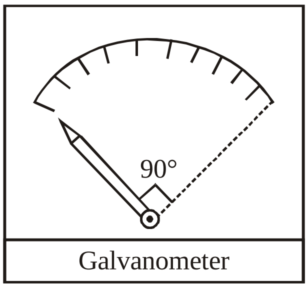
38. Three blocks each of mass m are kept on a wedge as shown. Coefficient of friction between each block and wedge is $\mu = 0.1$. Wedge is moved towards right with constant acceleration 'a', such that blocks are at rest with respect to wedge. Minimum possible value of 'a' for this to be possible is $\frac{10\alpha}{103}\, m/s^2$. Here α is an integer. Find α. (Strings are perfectly parallel to the slopes), (Use $g = 10 m/s^2$).



39. A particle projected from ground at some angle with vertical has displacement (w.r.t. point of projection) $\vec{S}_1 = (6\hat{i} + 8\hat{j})m$ and $\vec{S}_2 = \left(8\hat{i} + 6\hat{j}\right)m$ at some moment t_1 and t_2 respectively. If the range of the projectile is $\frac{37 \times n}{7} m$, then calculate 'n'. :-

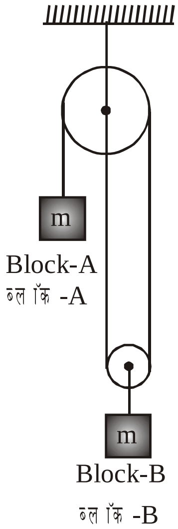
40. A particle of mass m = 1 kg lying at rest on x-axis, experiences a net force given by $F = x(3x - 2)$ Newton, where x is the x-coordinate of the particle in meters. The magnitude of minimum velocity in negative x-direction to be imparted to the particle placed at x = 4 meters such that it reaches the origin is $\sqrt{\frac{P}{27}}$ m/s. Find the value of $\frac{P}{1300}$.

41. In a moving coil galvanometer, a coil of area π cm^2 and 10 windings is used. Magnetic field strength applied on the coil is 1 Tesla and torsional stiffness of the torsional spring is 6×10^{-5} N.m/rad. A needle is welded with the coil. Due to limited space, the coil (or needle) can rotate only by 90º. For marking, the 90º space is equally divided into 10 parts as shown. Find the least count of this galvanometer.



(A) 1 mA (B) 2 mA (C) 3 mA (D) 4 mA

42. Both the blocks shown in figure have same mass ‘m’. All the pulley and strings are massless.



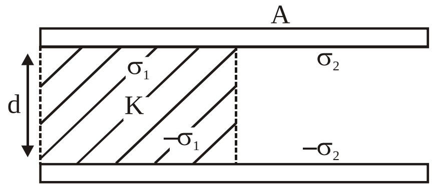
(A) Acceleration of block A is $\frac{2g}{5}$

(B) Acceleration of block A is $\frac{g}{5}$

(C) Acceleration of block B is $\frac{2g}{5}$

(D) Tension in the string attached with A is $\frac{6mg}{5}$

43. A parallel plate capacitor of area A and separation d is charged to potential difference V and removed from the charging source. A dielectric slab of constant K = 2, thickness d and area $\frac{A}{2}$ is inserted, as shown in the figure. Let σ_1 be free charge density at the conductor-dielectric surface and σ_2 be the free charge density at the conductor-vacuum surface. Then choose the **incorrect** option:



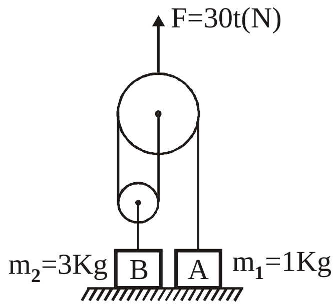
(A) The electric field inside the dielectric will be half of the electric field in the free space between the plates.

(B) The ratio $\frac{\sigma_1}{\sigma_2}$ is equal to $\frac{2}{1}$

(C) The new capacitance is $\frac{3 \in_0 A}{2d}$

(D) The new potential difference is $\frac{2}{3}$ V

44. Two blocks A and B having masses m_1 = 1 kg and m_2 = 3 kg are shown in figure. At t = 0 a force F = (30t) N is applied on pulley in vertically upward direction.(t is time in seconds, g = 10 m/s^2). Choose correct options –



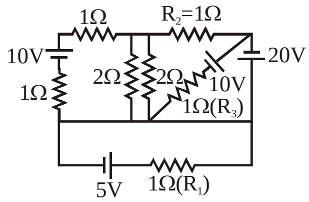
(A) A will loose contact at t = 2 sec and B will loose contact at t = 3 sec.

(B) A will loose contact at t = 1 sec and B will loose contact at t = 2 sec.

(C) Speed of A when B loose contact is 1.25 m/s

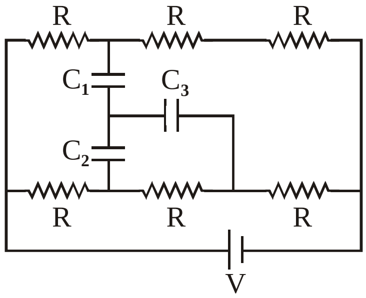
(D) Speed of A when B loose contact is 2.5 m/s

45. Find the sum of magnitude of currents through R_1, R_2 and R_3. All cells are ideal :



(A) 14 A (B) 30 A (C) 49 A (D) 52 A

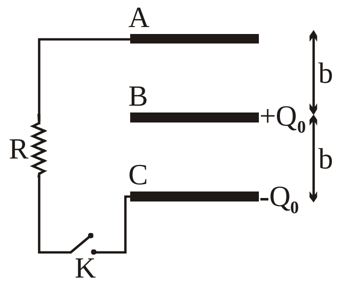
46. In the shown circuit, all three capacitor are identical and have capacitance C μF each. Each resistor has resistance of R Ω. An ideal cell of emf V = 108 volt is connected as shown. Then the magnitude of potential difference across capacitor C_3 in steady state is :



(A) 12 V (B) 24 V (C) 36 V (D) 48 V

Paragraph for Question Nos. 47 to 48

Three identical metal plates of area 'S' are at equal distances 'b' as shown. Initially metal plate A is uncharged, while metal plates B and C have respective charges $+Q_0$ and $-Q_0$ initially as shown. Metal plates A and C are connected by switch K through a resistor of resistance R. The key K is closed at time t = 0.



47. Then the magnitude of current in amperes through the resistor at any later time t is :

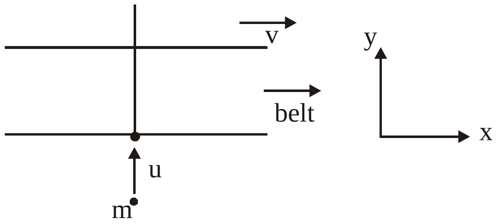
(A) $\dfrac{Q_0 b}{RS\varepsilon_0} e^{\frac{-bt}{RS\varepsilon_0}}$ (B) $\dfrac{Q_0 b}{RS\varepsilon_0} e^{\frac{-2bt}{RS\varepsilon_0}}$ (C) $\dfrac{Q_0 b}{2RS\varepsilon_0} e^{\frac{-2bt}{RS\varepsilon_0}}$ (D) $\dfrac{Q_0 b}{2RS\varepsilon_0} e^{\frac{-bt}{RS\varepsilon_0}}$

48. The total heat produced dissipiated by resistor of resistance R is :

(A) $\dfrac{Q_0^2 b}{4S \varepsilon_0}$ (B) $\dfrac{Q_0^2 b}{8S \varepsilon_0}$ (C) $\dfrac{Q_0^2 b}{2S \varepsilon_0}$ (D) $\dfrac{3 Q_0^2 b}{4S \varepsilon_0}$

Paragraph for Questions no. 49 & 50

A conveyer belt is moving with speed v in x-direction as shown in figure. A particle of mass m is projected on this belt with speed u(w.r.t. ground) in y-direction. Coefficient of friction between particle & belt is μ. The particle touches the belt at the origin of a fixed coordinate system and remains on the belt.



49. Co-ordinates of particle when it stop with respect to frame fixed on belt, is

(A) $\left(\frac{v\sqrt{v^2+u^2}}{2\mu g}, \frac{u\sqrt{v^2+u^2}}{2\mu g}\right)$
(B) $\left(-\frac{v\sqrt{v^2+u^2}}{2\mu g}, \frac{u\sqrt{v^2+u^2}}{2\mu g}\right)$
(C) $\left(-\frac{v\sqrt{v^2+u^2}}{\mu g}, +\frac{u\sqrt{v^2+u^2}}{\mu g}\right)$
(D) $\left(-\frac{v\sqrt{v^2+u^2}}{4\mu g}, +\frac{u\sqrt{v^2+u^2}}{4\mu g}\right)$

50. Coordinates of particle when it stop with respect to belt in the frame fixed on ground, is

(A) $\left(\frac{v\sqrt{v^2+u^2}}{\mu g}, \frac{u\sqrt{v^2+u^2}}{\mu g}\right)$
(B) $\left(-\frac{u\sqrt{v^2+u^2}}{2\mu g}, \frac{v\sqrt{v^2+u^2}}{2\mu g}\right)$
(C) $\left(\frac{v\sqrt{v^2+u^2}}{2\mu g}, \frac{u\sqrt{v^2+u^2}}{2\mu g}\right)$
(D) $\left(\frac{u\sqrt{v^2+u^2}}{4\mu g}, \frac{v\sqrt{v^2+u^2}}{4\mu g}\right)$

ANSWER KEY TO DPP # 02 (JEE-ADVANCED)

1.	(A,B,C,D)	**2.**	(A,C)	**3.**	(A,C)	**4.**	(A,B,D)	**5.**	(B,D)
6.	(A,B,D)	**7.**	(A,B,D)	**8.**	(B,D)	**9.**	(A,C,D)	**10.**	(A,B,C)
11.	(B,C,D)	**12.**	(A,D)	**13.**	(A,B,C,D)	**14.**	(B)	**15.**	(B,C)
16.	(A,B,C)	**17.**	(A,C)	**18**	(A,C,D)	**19.**	(A,C)	**20.**	(B,C)
21.	03	**22.**	–09	**23.**	81	**24.**	12	**25.**	40
26.	04	**27.**	05	**28.**	03	**29.**	03	**30.**	10
31.	04	**32.**	–07.50	**33.**	06	**34.**	05	**35.**	02
36.	09	**37.**	–08	**38.**	20	**39.**	02	**40.**	02
41.	(C)	**42.**	(A)	**43.**	(A)	**44.**	(C)	**45.**	(C)
46.	(B)	**47.**	(B)	**48.**	(A)	**49.**	(B)	**50.**	(C)

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